

4. Model Scoring and Evaluation

4.1 Performance Metrics Comparison

Due to the severe class imbalance in the Toronto KSI dataset (where non-fatal injuries vastly outnumber fatalities), evaluating our models based solely on raw Accuracy is highly misleading. A model could achieve 86% accuracy simply by guessing "Non-Fatal" for every single record.

In fact, our **Random Forest** model is a textbook demonstration of this exact trap: it achieved the highest overall accuracy (86.63%) paired with an abysmal recall (5.86%). It is essentially acting as a majority-class guesser. Because of this, we prioritized **Recall** (catching as many true fatalities as possible for public safety) and the **F1-Score** (balancing false alarms with missed fatalities), alongside **ROC-AUC** to measure the overall class separation capability.

Below are the test-set results for the models utilizing our SMOTE-balanced training pipeline:

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Neural Network (MLP)	83.44%	37.97%	30.91%	0.341	0.728
Support Vector Machine	77.79%	28.49%	40.00%	0.333	0.684
Logistic Regression	65.01%	20.79%	54.34%	0.301	0.681
Decision Tree	79.58%	26.36%	26.46%	0.264	0.573
Random Forest	86.63%	70.73%	5.86%	0.108	0.720

4.2 Receiver Operating Characteristic (ROC) Analysis

To further visualize the diagnostic ability of our classifiers, we plotted the ROC curves. The ROC curve illustrates the trade-off between the True Positive Rate (Recall) and the False Positive Rate across different probability thresholds.

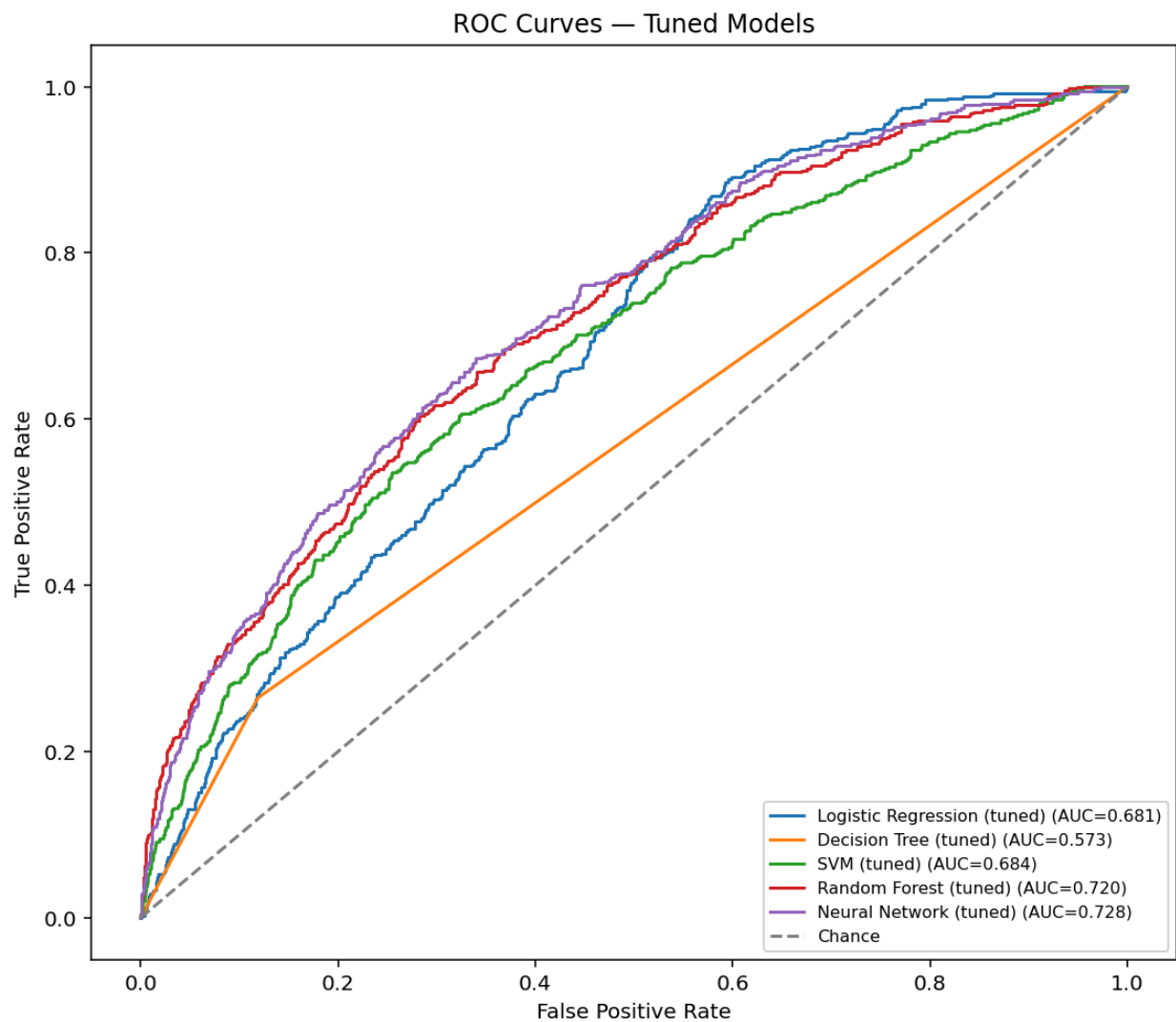


Figure 1: ROC Curves comparing diagnostic capability across tuned models.

4.3 Confusion Matrices Analysis

To understand exactly where our models are succeeding and failing, we generated confusion matrices for each algorithm, evaluating their performance against the 3,575 records in our test set (3,080 Non-Fatal vs. 495 Fatal).

- **Neural Network (Tuned):** Correctly identified 153 fatalities, missed 342 (False Negatives), and triggered 250 False Positives.
- **Support Vector Machine:** Correctly identified 198 fatalities, missed 297, and triggered 497 False Positives.
- **Logistic Regression:** Correctly identified 269 fatalities, missed 226, but triggered a massive 1,025 False Positives.

- **Decision Tree:** Correctly identified 131 fatalities, missed 364, and triggered 366 False Positives.
- **Random Forest:** Correctly identified only 29 fatalities, missed 466, and triggered 12 False Positives.

4.4 Final Model Selection and Recommendation

After extensive hyperparameter tuning using GridSearchCV and RandomizedSearchCV, we recommend the **Neural Network (MLPClassifier)** as the primary model for deployment, with **Logistic Regression** serving as a highly viable alternative depending on the specific needs of the Toronto Police.

Our justification for this selection is based on the following:

1. **Best Overall Balance (Highest F1):** The Neural Network achieved the highest overall F1-Score (0.341). In a real-world public safety tool, missing a fatal intersection (False Negative) costs lives, but "alarm fatigue" from too many False Positives causes the police to ignore the system. The Neural Network achieved the best mathematical balance of these two extremes. The Support Vector Machine (SVM) performed competitively (F1: 0.333, Recall: 40.0%), but ultimately offered no advantage over the Neural Network on either metric.
2. **Highest Discriminative Power (ROC-AUC):** The Neural Network achieved a highly competitive Area Under the Curve of 0.728, indicating it possesses a strong overall ability to correctly distinguish between a fatal and non-fatal collision across all probability thresholds.
3. **Threshold Calibration Potential:** Random Forest's near-competitive AUC (0.720) suggests its poor recall may be a threshold-calibration issue rather than a fundamental ranking weakness. This is worth revisiting with a lower probability cutoff in future work.
4. **The "Recall" Trade-off (Why we also recommend Logistic Regression):** It is crucial to note that if the city's only goal is to cast the widest net possible to catch every potential fatality (regardless of false alarms), the Logistic Regression model actually performed best in that specific metric, achieving a Recall of 54.34%. However, because its precision was so low (20.79%), the Neural Network remains our official recommendation for a balanced, deployable web API.

Feature Importance Insights

As visualized in our Feature Importance findings below, the tree-based ensemble confirmed that geographic location (Latitude/Longitude), time of day, and vehicle type are the strongest predictors of traffic fatalities, providing actionable insights for city safety planners.

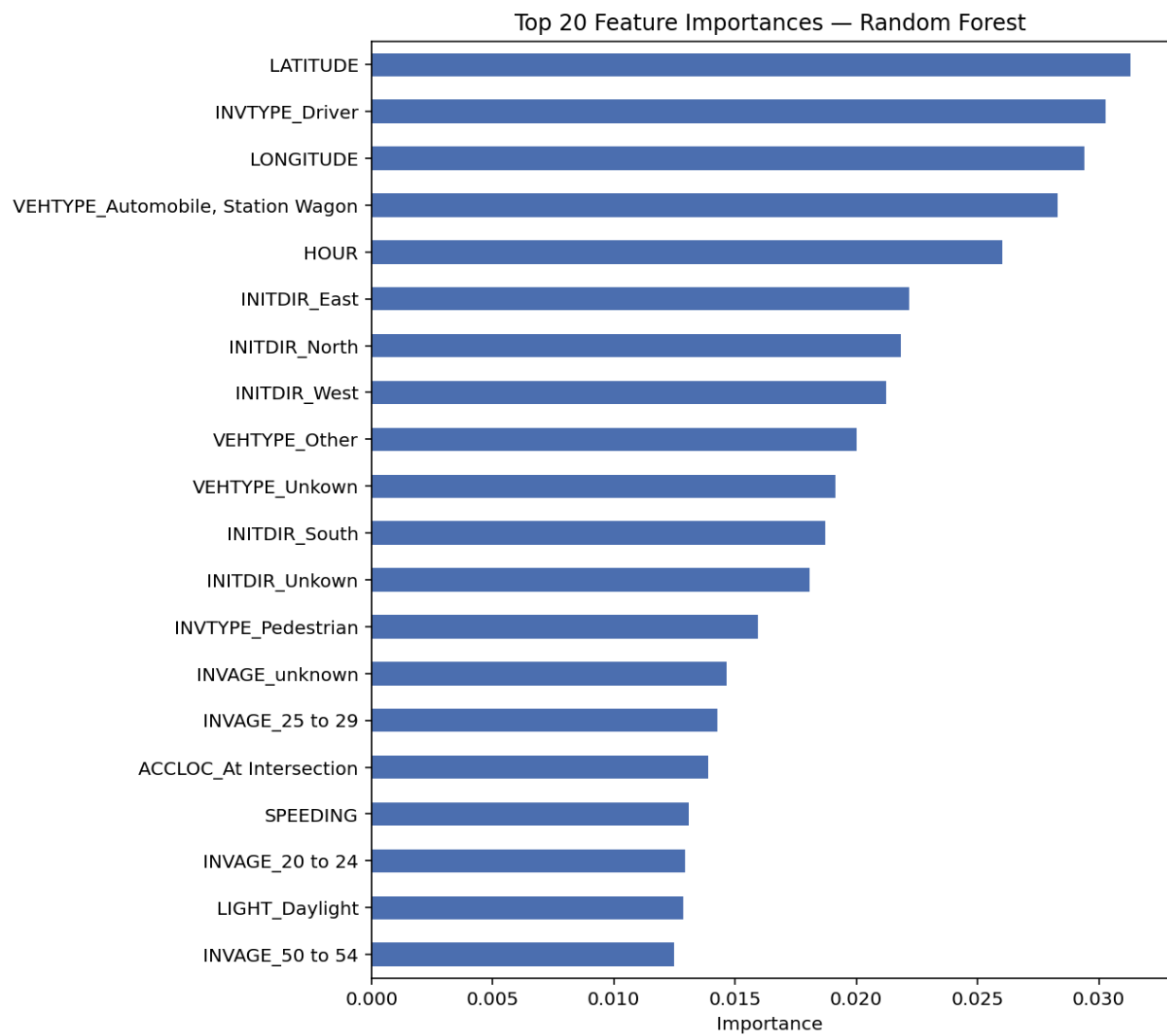


Figure 2: Top 20 feature importances identified by the Random Forest model.